

AutoBench

Architecture & Design Overview

A pure-Python, HTTP-only service that deploys and evaluates agent benchmarks across multiple cluster-specific Rossoctl instances.

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Agenda

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1. Overview & Key Design Decisions

What it is

Deploys a benchmark's MCP tool + A2A agent, runs the evaluation, reads results, and exports them — all over HTTP.

Objectives

- Automate benchmarking lifecycle operations
- Cross-user/cluster sharing of benchmark run results
- Easy to configure in-cluster & cross-cluster benchmark runs
- Asynchronous parallel benchmark runs
- Secure, scalable, auditable & resilient

3 benchmarks

- gsm8k — single-turn: one prompt in, one answer out, no dialogue partner
- tau2 — multi-turn: a server-side user simulator LLM plays the customer
- appworld — long-horizon: many API calls across simulated apps

Client contract

- Point at a hostname, send Authorization: Bearer <caller JWT> — same from kind to prod

Key design decisions

Pure-Python, HTTPS-only to Rossoctl

No shell-out: smaller image, no injection surface, structured errors, testable.

Two-token auth model

Caller JWT attributes + routes only (never forwarded); Service mints its own benchmarker (ROPC) token to Rossoctl.

iss is the trust anchor

The JWT issuer selects the per-instance config — no separate instance argument, so no confused-deputy escape.

Per-request ROPC login

Fresh Service token every request — no expiry handling.

iss-keyed per-instance config

One file per issuer: Rossoctl URL, benchmarker cred, MLflow/S3, optional Keycloak backchannel + workload route templates.

Enact only what the API allows

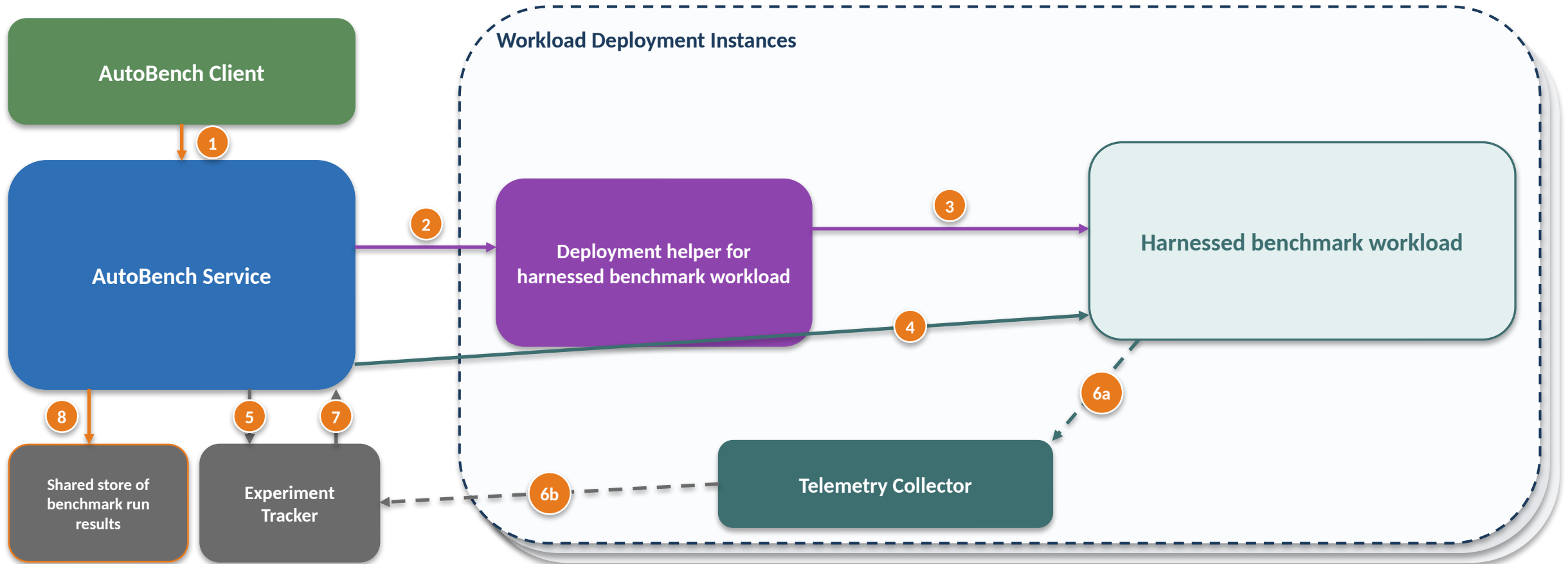
Deploy/run/report/export — now including AuthBridge plugin presets (layer-3). Only workload Secrets remain out-of-band: prechecked and reported (424), never silently ignored.

MLflow with Service; S3 in the cloud

MLflow is co-located with the Service (per-service traces it emits + reads), not in the workload cluster; S3 is an external cloud service — the shared cross-service sink.

2. Auto-Benchmarking Design Motif

One client, one Service, N workload deployments — and every result ends up in one place

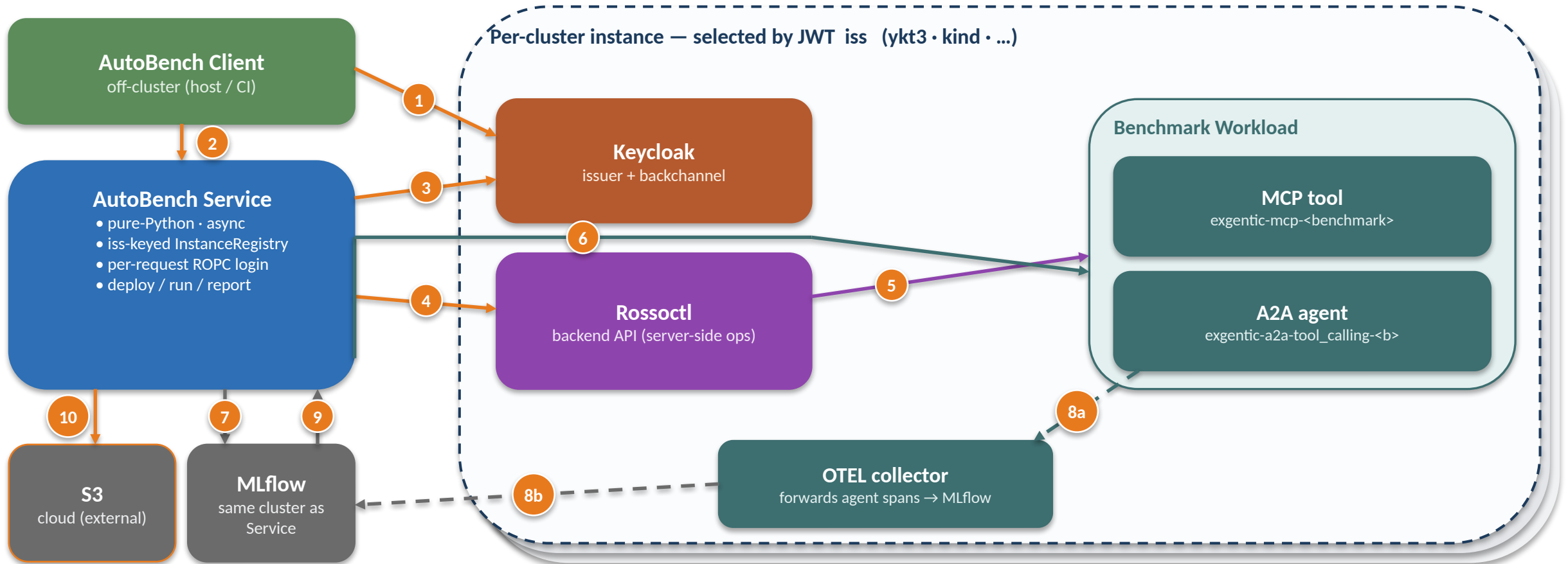


- 1 Client → Service
- 2 Service → deployment helper: create the harnessed workload
- 3 helper → harnessed workload
- 4 Service → harnessed workload: run and monitor benchmark execution

- 5 Service → Experiment Tracker: emit the trace
- 6a workload → Telemetry Collector · 6b collector → Tracker
- 7 Experiment Tracker → Service: read the records back
- 8 Service → shared store: export the run's artifacts

3. Implementation Architecture: Kubernetes-Based Benchmark Workloads

Relations between client, service, cluster-specific Keycloak / Rossoctl / workload, MLflow & S3

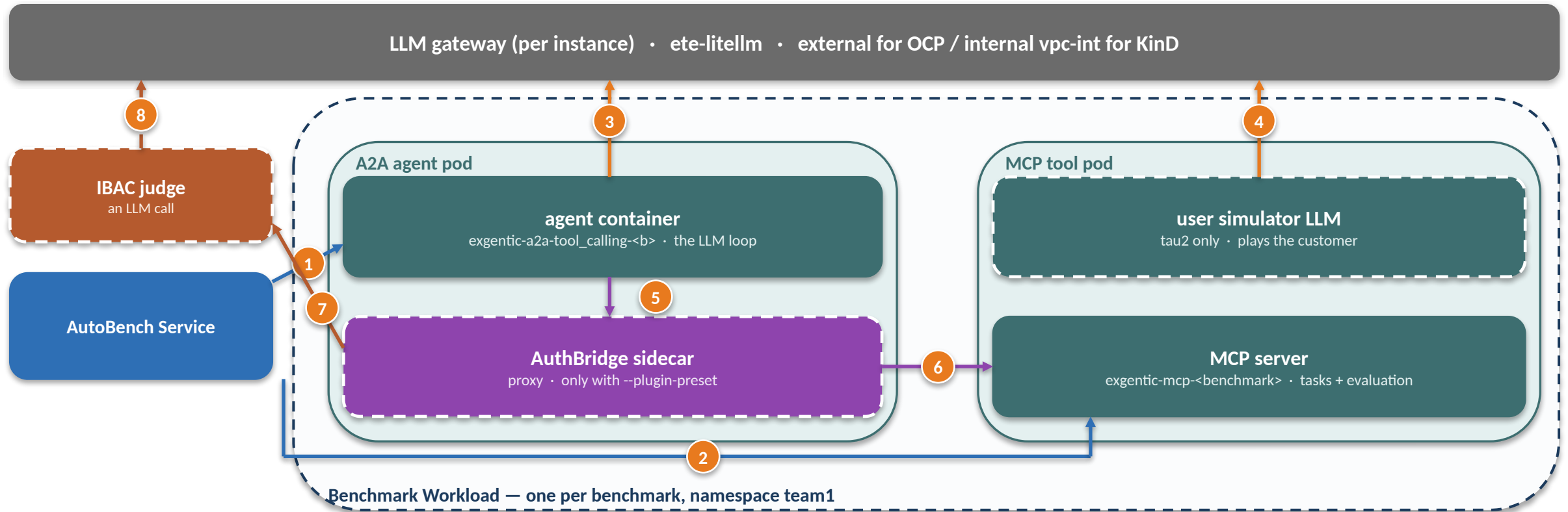


- 1 Client logs in to Keycloak (ROPC) → caller JWT
- 2 Client → Service: Authorization: Bearer <caller JWT>
- 3 Service validates JWT (JWKS) + mints its own benchmarker token (ROPC, via backchannel)
- 4 Service → Rossoctl: deploy / list agents + tools
- 5 Rossoctl creates the MCP tool + A2A agent in-cluster

- 6 Service drives the MCP session + the A2A call itself; the agent only issues execute_tool inside that session
- 7 Service emits the Agent.Session trace → MLflow (first)
- 8a A2A agent → OTEL collector · 8b collector → MLflow (optional workload spans — off by default)
- 9 Service reads back Agent.Session traces (MLflow)
- 10 Service exports run.json / report.* (S3)

4. Architecture with Workload Specific Components

Inside the Benchmark Workload — what every benchmark has, and what only some of them add



Which components a benchmark actually gets

benchmark	MCP tool image	LLMs per task	tool calls
gsm8k	exgentic-mcp-gsm8k	1 (agent)	~1
tau2	exgentic-mcp-tau2	2 (+ user simulator)	~11
appworld	exgentic-mcp-appworld	1 (agent)	~13
+ any preset	unchanged	+1 IBAC judge call per action	—

- Service → agent: send_prompt, once per task
 - Service → MCP server: list_tasks, create_session, evaluate_session, delete_session (the verdict)
 - agent → gateway: 1 probe + N real chat calls
 - tau2 only: the user simulator is a 2nd LLM
 - with a preset: the agent's MCP calls go via the sidecar
 - sidecar → MCP server, once authorized
 - sidecar → judge, per isAction call
 - the judge is itself an LLM call
- Dashed outline = present only in some configurations.
No preset: 5 + 6 become one direct agent → MCP call.

5. The Two-Token Auth Model

Why the caller's token is never forwarded upstream

Caller JWT (inbound)

Presented by the client as Authorization: Bearer.

- Used ONLY to attribute (preferred_username) and route (iss → instance)
- Signature validated via the issuer's JWKS
- aud / exp deliberately lenient in the dev/ops context
- **NEVER forwarded to Rossoctl**

benchmarker is special:

- A caller JWT with preferred_username == benchmarker authorizes config ops (GET/PUT /config)

Service token (outbound)

Minted by the Service per request via ROPC (password grant) using the instance's benchmarker credential.

- Always fresh → no expiry handling
- Presented to Rossoctl for all cluster-facing ops
- **The Service acts under its OWN identity, not the caller's**

Backchannel split (iss ≠ dialed host):

- When the iss host is unreachable (in-cluster kind), JWKS + token come from a configured backchannel URL; iss is matched, never dialed

Implication: at the Rossoctl / cluster layer every action appears as the Service's identity — so per-user attribution & audit live in the Service, keyed on (iss, preferred_username).

6. Benchmark Catalog & Run Lifecycle

Catalog (static registry)

gsm8k

single-turn · needs hf-secret + openai-secret

tau2

multi-turn · user-simulator LLM runs server-side in the MCP pod · raise timeout

appworld

long-horizon across simulated apps · runs e2e · raise task_timeout_seconds

Lifecycle (REST)

POST /benchmarks/{n}/deploy

create MCP tool + A2A agent → 201

GET /benchmarks/{n}/status

poll until tool_ready & agent_ready

POST /benchmarks/{n}/runs

precheck → 202 run_id (409 not deployed · 424 secret missing)

GET .../runs/{id}

poll pending → running → succeeded / failed

GET .../runs/{id}/report

MLflow records + artifacts (409 if MLflow unset)

download from S3

run.json · report.* · token_report.* · span_report.* · manifest.json

7.1 The Three Benchmarks — a Difficulty Ladder

Not interchangeable suites: each costs ~an order of magnitude more than the last

	gsm8k	tau2	appworld
What it tests	multi-step arithmetic	multi-turn dialogue + tools	long-horizon app automation
Tasks measured	172 (all clean)	60 (all clean)	39 (all clean)
Pass rate	0.99	0.82	0.00
Input tokens / task	341	82,957	254,243
Output tokens / task	192	1,969	24,157
LLM calls / task	2.1	11.1	28.3
Tool calls / task	1.1	11.2	13.6
Median task latency	7.7 s	97 s	290 s
Slowest task seen	29 s	181 s	596 s
Model we use	gpt-5-mini	claude-sonnet-5	gemini-2.5-pro
Task pool	8.5K (HuggingFace)	114 (retail domain)	grouped scenarios

The scale gap is the headline: a tau2 task costs ~245x the input tokens of a gsm8k task, an appworld task ~745x. A 50-task gsm8k run is minutes; a 20-task appworld run is half an hour and millions of tokens. Budget by benchmark, not by task count.

7.2 What Each Benchmark Stresses

Why all three are in the matrix, and what a result from each does and does not tell you

gsm8k — the canary

- One prompt in, one answer out — no dialogue partner.
- ~1 real LLM call + 1 tool call: the simplest agentic loop.
- Stresses almost nothing about the platform — which is the point.
- A failure here means INFRASTRUCTURE: deploy, auth, LLM reach, telemetry.
- Saturated at ~1.0, so it cannot discriminate models. Never read it as one.
- Deterministic enough that task 0 costs 320 input tokens on every cluster — we use that to prove two environments are comparable.

tau2 — the discriminator

- Multi-turn: a server-side USER SIMULATOR LLM plays the customer.
- Two models talk to each other, plus ~11 tool calls per task.
- Domain is retail (114 tasks) — the library default, never recorded in artifacts.
- The only leg where model choice dominates: 0.1 with gpt-5-mini vs 0.9 with claude-sonnet-5 on the SAME 10 tasks.
- Episodes are nondeterministic: across 6 samples of the same 10 tasks, 4 flip.
- At n=10 one task moves pass_rate by 0.10 — it cannot resolve less than that.

appworld — the stress test

- Realistic chores across simulated apps: discover APIs, chain many calls.
- ~28 LLM calls, ~14 tool calls, ~254k input, ~5 min per task.
- Evaluation is PROGRAMMATIC unit tests over final app state — no LLM judge.
- Pass rate 0.0 is honest, not broken: runs complete and tokens record — the agent just does not finish the job.
- ~92% of tasks call finish: it believes it is done and the assertions disagree. No verification pass.
- Its job here is pipeline stress: long contexts, big traces, real timeouts.

If you want to ...

check a cluster / deploy / auth / telemetry path works

exercise concurrency and volume cheaply

compare models meaningfully

stress long contexts, long tasks, timeouts

use

gsm8k, 1–10 tasks

gsm8k, 50 tasks at max_parallel_sessions=4

tau2 — it discriminates; gsm8k saturates at ~1.0

appworld

7.3 Reading the Numbers — Five Things That Mislead

Every one of these cost us a wrong conclusion first

1. $\text{pass_rate} = \text{evaluated_pass} / \text{total}$

A task that ERRORS before evaluation counts as not passed, so a low rate can mean “failed the task” or “never got judged”. Check the error column too.

2. The llm column overcounts real calls by one

Every task issues an extra $\text{max_tokens}=1$ capability probe, counted as a chat span. $\text{llm}=2$ on gsm8k means ONE real call.

3. Implausibly small tokens = lost telemetry, and $\text{tokens} == 0$ will not catch it

When a usage-bearing span is lost, what survives is the probe — whose own usage is 0/0 on reasoning models but 8/1 on claude-sonnet-5 and 1/0 on gemini. Use the structural test: $\text{llm} \leq 1$ with $\text{tool} \geq 2$ is impossible.

4. Output varies MORE than input, in most runs

Measured $\text{OUT CV} > \text{IN CV}$ in 27 of 33 legs. gsm8k's prompt is near-constant while answer length swings; only long-horizon appworld inverts it. Do not infer a direction — read the CV.

5. Task selection is deterministic

A run takes the first max_tasks tasks, so the same task_id is the same task across runs and clusters, and a smaller run is a prefix of a larger one. That is what makes cross-platform comparison like-for-like — six legs matched to the byte.

8.1 The Canonical 12-Run Matrix

One fixed set of 12 request bodies — the same on every platform, every version

#	benchmark	tasks	parallel	what it varies
1	gsm8k	1	1	baseline — the smoke test
2	gsm8k	10	1	volume, still serial
3	gsm8k	50	4	volume + concurrency
4	gsm8k	5	4	model swap → Azure/gpt-4.1
5	gsm8k	5	4	AuthBridge preset: auth-only
6	gsm8k	5	4	AuthBridge preset: ibac-only
7	gsm8k	5	4	AuthBridge preset: full (enforce)
8	gsm8k	5	4	full + per-plugin override ibac:observe
9	tau2	10	1	multi-turn + user simulator
10	tau2	20	4	multi-turn under concurrency
11	appworld	5	1	long-horizon, gemini-2.5-pro
12	appworld	20	4	long-horizon under concurrency

Why a FIXED matrix: task selection is deterministic, so the same leg run anywhere executes the same tasks in the same order. That is what makes a difference attributable to the platform rather than to the workload — and it is why every leg deploys fresh, since reusing a warm agent silently loses telemetry. Driven by one command (reference/run-12.py, ~1h50m); the three generated documents in results/ derive every number from the mirrored artifacts.

8.2 What the 12 Runs Measured

v1.26 on OpenShift (Service on ykt3, workloads on ykt2) — 136 tasks, all 12 legs succeeded

#	bench	pass	wall	input tokens
1	gsm8k	1.00	4 s	320
2	gsm8k	1.00	63 s	3,166
3	gsm8k	1.00	89 s	15,684
4	gsm8k	1.00	11 s	4,213
5	gsm8k	1.00	28 s	1,564
6	gsm8k	1.00	32 s	1,564
7	gsm8k	1.00	25 s	1,564
8	gsm8k	1.00	30 s	1,564
9	tau2	0.80	802 s	863,927
10	tau2	0.85	483 s	1,656,724
11	appworld	0.00	1619 s	868,819
12	appworld	0.00	1804 s	3,598,730

All eight gsm8k legs pass 1.00

Including #4's gpt-4.1 swap, which cost one task of five on the previous matrix. The four AuthBridge presets do not change the result — only the latency.

tau2 is the only leg that moves

0.80 and 0.85 here, 0.90 and 0.75 on KinD. Across samples of the same 10 tasks some flip: at n=10 one task is worth 0.10, so it cannot resolve less.

appworld 0.00 is the honest result

Runs complete, tokens record, evaluation says the goal was not met. ~92% of tasks self-declare finish.

Token attribution complete: 0 of 136 rows lost

An earlier matrix had 15 damaged rows that a tokens == 0 check reported as clean. Fresh deploy per leg, plus a structural detector.

7.0M input / 513k output tokens

over 4,990 s of wall time — appworld alone is 64% of the input, on 20 of 136 rows.

9.1 OpenShift vs KinD — Like-for-Like

Same 12 request bodies, same Service version, verified-identical instance config

#	bench	pass OCP	pass KinD	in OCP	in KinD	
1	gsm8k	1.00	1.00	320	320	identical
2	gsm8k	1.00	1.00	3,166	3,166	identical
3	gsm8k	1.00	1.00	15,684	15,684	identical
4	gsm8k	1.00	0.80	4,213	3,642	
5	gsm8k	1.00	1.00	1,564	1,564	identical
6	gsm8k	1.00	1.00	1,564	1,564	identical
7	gsm8k	1.00	1.00	1,564	1,564	identical
8	gsm8k	1.00	1.00	1,564	1,564	identical
9	tau2	0.80	0.90	863,927	910,744	
10	tau2	0.85	0.75	1,656,724	1,546,050	
11	appworld	0.00	0.00	868,819	1,071,356	
12	appworld	0.00	0.00	3,598,730	4,376,562	

9 of 12 pass rates identical

The three that differ are #4 (one gsm8k task of five) and both tau2 legs — and tau2 is the leg we know flips run to run on either platform.

7 legs byte-identical on input tokens

320 / 3,166 / 15,684 / 1,564 x4. Deterministic task selection plus the same model means identical work — the strongest available proof this is a like-for-like comparison, not merely a similar one.

Wall time: 4,990 s OCP vs 5,986 s KinD

The single node is 20% slower overall, and individual legs differ by up to 3.4x in either direction — appworld dominates the total.

0 lost-attribution rows on both sides

136 OCP rows, 135 KinD rows.

So the platform is not the variable

Where the two differ it traces to episode nondeterminism (tau2), small samples (#4 = one task), or appworld task timeouts — not to OpenShift vs KinD.

9.2 What AuthBridge Costs

Legs #3 and #5–8 ran byte-identical work, so latency differences are the plugin config

component	cost	measured from
Sidecar present at all	+12.6 s / task (OCP)	flat-ish across all four presets
	+1.0 s / task (KinD)	same legs, single node
Tool call, sidecar but unjudged	+1.56 s (OCP)	auth-only; +0 ms on KinD
Tool call, judged	not isolable	no leg judged all 5 calls

median span	OCP	KinD	ratio
connect_mcp, no sidecar	424 ms	79 ms	5.4x
connect_mcp with sidecar	3200 ms	131 ms	24.4x
tool call, unjudged	1657 ms	11 ms	150.6x
tool call, judged	4689 ms	1306 ms	3.6x

The judge is itself an LLM call

The sidecar POSTs to a chat-completions endpoint per authorized action, which is why it costs seconds and why its latency is so variable.

Most of the OCP figure is topology, not plugins

The judged-call cost is broadly comparable across platforms (3.6x) because it is dominated by a shared external gateway. The fixed cost is 13x apart because this OCP matrix is cross-cluster — token exchange over external routes. Do NOT quote +12.6 s as “the cost of AuthBridge”; the single-node figure (~+1.0 s fixed) is the better estimate of the intrinsic cost.

Presets cannot be ranked from these numbers

NOT ONE of the six IBAC legs authorized all of its tool calls: OCP judged 4/5, 3/5, 4/5 and KinD 3/5, 2/5, 3/5. A leg that judged fewer calls shows a lower median — a mixture, not a saving, which is why the judged-call cost cannot be isolated at all from this matrix. Decision caching under concurrency, or a fail-open hole? Still OPEN; the test is one ibac-only leg at parallelism 1.

Latency only

The sidecar's CPU and memory cost is unmeasured — every infra field in these runs is 0.0.

10. What the Service Can & Cannot Enact

The HTTP-only boundary, made explicit

✓ Enacts over HTTP

- Deploy MCP tool + A2A agent (CPU/mem, image, env)
- Deploy-time model swap (per-experiment agent)
- authbridge_enabled → inject the sidecar (layer-2)
- plugin_preset / plugins / on_error → AuthBridge layer-3
→ pluginPreset/plugins/onError; operator renders ConfigMap
- Run benchmark sessions; collect pass/fail + latency
- Emit + read MLflow traces; export to S3
- Service-owned config: MLflow + S3 via PUT /config
benchmarker only

✗ Out-of-band (reports, doesn't do)

- Cluster Secrets (hf-secret, openai-secret)
operator provisions; run precheck returns 424 naming it
- AuthBridge CLUSTER config — ibac judgeEndpoint / judgeModel
in the platform-config ConfigMap, not the agent API
- Any cluster-level API call — Rossoctl does these server-side

Principle:

- Never silently ignore an un-enactable request —
precheck (424) or reject (422), with an actionable reason

Cluster-agnostic by construction: works on kind / vanilla k8s / OpenShift; cross-cluster runs use per-instance route templates + a reachable internal issuer.